



Unleashing Open-Source Potential in SAS: The PharmaForest Ecosystem (Proc PharmaForest)

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Agenda



- Open-Source in Pharmaceutical
- What is PharmaForest?
- SAS Packages Framework (SPF)
- Growth and Expectations
- Organizational Participation
- Diverse Programming Environment
- Conclusion
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Open-Source in Pharmaceutical

Open-Source Surge in Pharmaceutical



- Open-source communities (example: pharmaverse) encourage industry collaboration showing how unified solutions, reusable codes, and standardized workflows for statistical programming and data analysis can reduce duplication of efforts and gain increased harmonization to the way we work across the industry.
- Key Drivers:
 - **Efficiency:** Timelines are tightening; redundant programming and validation are costly and time consuming.
 - **Collaboration:** Cross-industry alignment on standards and methods (e.g., CDISC compliant submissions).
 - **Innovation:** Rapid iteration with community peer review (vs. isolated development).

How are SAS Users changing?



- The open-source R ecosystem is accelerating collaboration in drug development and can reduce redundant effort.
- The open-source mindset is spreading beyond R (e.g., shared standards, reusable components, open validation practices).
- This is encouraging some SAS professionals to share code and patterns more openly as well.
- Some SAS programmers are increasingly exploring R to benefit from open tooling and community support.
- Others continue using SAS but adopt open-source-inspired practices (reuse, transparency, contribution) in SAS or hybrid workflows.
- There is a need for a unified repository of SAS packages* to enable community collaboration and sustainability.

What is PharmaForest?

What is PharmaForest?

- PharmaForest is a collaborative, **Open-Source** community to expand/maintain a repository of SAS packages for pharmaceutical use cases.
- The initiative was established in **2025**.
- Operates within **PHUSE Data Visualization & Open-Source Technology** working group context.
- As of April 2026, PharmaForest has **45 packages** and still rapidly growing in number of packages.
- PharmaForest is positioned as a unified repository of SAS packages for users in pharmaceutical and intends to serve a similar role as it's being done for R.



github.com/pharmaforest



PharmaForest

README.md

PharmaForest (we have already 45 packages 🦊🦊🦊)

A collaborative repository of SAS packages for pharmaceutical industry, powered by PHUSE Japan OST (Open Source Technology) WG members—Hiroki Yamanobe, Yutaka Morioka, and Ryo Nakaya—showcasing the list of available packages below. [Click the QR code to visit gallery page.](#)



[PharmaForest Gallery](#)

-Goal-

Through SAS Packages, we want to actively encourage sharing of SAS know-how that has often stayed within individuals. By doing this, we aim to build up collective knowledge, boost productivity, ensure quality through standardization, and energize our community.

-Priority-

Our first priority is to share openly—and get others to share as well—so that more people can join in. On that basis, we'll work on improving quality, driving standardization, and creating long-term value.

<https://github.com/pharmaforest>

PharmaForest Objectives



- Promote **sharing** of SAS expertise and resources that would serve the SAS community.
- **Consolidate** know-how into a unified hub to improve awareness and encourage standardization/quality.
- **Expand** participation to build a sustainable community.
- Promote open, collaborative sharing to **reduce** siloed SAS statistical program development, while remaining mindful of intellectual property boundaries.

PharmaForest Packages Overview



PharmaForest currently hosts **45 plus SAS packages**

Packages

#	Package	#	Package	#	Package	#	Package
1	OncoPlotter (PharmaForest original package)	16	maxims4sas (PharmaForest original package)	31	basketMiningKit	46	SASDoGs
2	sashash	17	laglead	32	datacut_sas	47	
3	sas_dataset_json	18	ctcae5_grader	33	retro_spell_check	48	
4	SASPACer	19	love_matrix	34	sas_dataset_toon	49	
5	misc (PharmaForest original package)	20	SASPACerShiny (R Package)	35	defensivechk	50	
6	SAS FAKER	21	change_tracer_dog	36	GetCDISCCodelist	51	
7	SASLogChecker	22	yaml_writer	37	kpi_viz_toolbox	52	
8	sas_compare	23	Adamski (PharmaForest original package)	38	outlier_screening	53	
9	rtfCreator	24	RWDExpress	39	sARDen	54	
10	shutter_chance	25	freqsum	40	sARDenX	55	
11	devil (PharmaForest original package)	26	qc_cat_referee	41	sARDenMTS	56	
12	xattr_kit	27	SASToolbox	42	corr	57	
13	cloak	28	trial_flow	43	sas2typst_report	58	
14	vis_review_kit	29	valivali (PharmaForest original package)	44	dataset_maker	59	
15	odstable_layoutshowcase	30	mergex	45	sashighlight	60	



SAS Packages Framework (SPF)

SAS Packages Framework (SPF)



- SPF is a collection of core SAS macros developed and maintained under the MIT license*.
- Enables a foundation for a modular component system in SAS, similar in spirit to R/Python packages.
- Supports creating, distributing, installing, and using SAS packages with all required components bundled together.
- PharmaForest is built on SPF, bringing modularity and standardized sharing into SAS.



https://github.com/yabwon/SAS_PACKAGES

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SAS Packages Framework (SPF) - Advantages



- Both SASAUTOS and compiled catalogs are mainly dedicated to macros (the latter also for formats).
- A SAS Package provides a standardized framework for organizing and distributing reusable code and assets, including macros, user-defined functions, formats, and other related components, within a single package.
- Packages can be shared among different operating systems, keeping easy access to source code for preview.
- Packages combines all necessary codes for proper execution into one file and allows to maintain an order of execution too (designed-by-developer order).
- A package binds all its code under version number with the timestamp and provides a unique SHA256* digest that facilitates easy verification or package tampering.
- Packages provide easily accessible documentation that can be printed in the SAS session log.
- SAS packages, allow deployment across every SAS environment (Base SAS, SAS9 servers with Enterprise Guide, SAS Viya with SAS Studio, and even the latest SAS Workbench with Visual Studio Code).

PharmaForest & SAS Packages Framework (SPF)



PharmaForest uses SPF as a foundation for building self-contained SAS packages.

- Manually create source files and folders and run %generatePackage
- Use [SASPACer](#)
- Use [SASPACerShiny](#) and run %generatePackage
- Ask [Oba-chan](#) (SAS Package Granny) to cook source files and folders and run %generatePackage
- Use [Agent skills for SPF \(spf-skills\)](#) and run %generatePackage

Steps to generate SAS Package:

1. Set-up SPF Framework
filename packages "¥path¥to¥your¥packages";
2. Enable SPF Framework
%include packages(SPFinit.sas)
3. Generate Package
*%generatePackage(filesLocation =
/path/to/package/content, markdownDoc = 1)*

[SPF documentation](#)

PharmaForest & SAS Packages Framework (SPF)



SAS Package Implementation

Typical workflow (example package name **xxx**):

- `%installPackage(xxx, mirror=pharmaforest)`
- `%loadPackage(xxx)`

Note: "PharmaForest packages are distributed and consumed using SPF."

Growth and Expectations

Growth and Expectations



- Initiated in **June 2025 PharmaForest's** rapid early growth was intentional to build visibility (40+ packages in ~6 months).
- It has evolved from a Japan-local PHUSE working group activity to a more global effort.
- PharmaForest's "dual strategy":
 - Pursue high-quality, potentially regulatory-compliant packages.
 - Also encourages broad publishing to lower participation barriers and grow adoption.
- Looking for sustainability, requiring maintenance capacity (updates, fixes) via broader collaboration.
- Next focus phase: Governance and validation.

Organizational Participation

Organizational Participation



- Currently mostly individual contributors; company-level involvement not formalized.
- Need clarity on:
 - Scope of what can be shared (IP boundaries).
 - Contribution/review/maintenance practices aligned to corporate governance/compliance.
- Position PharmaForest as a platform to share open tools across organizations (distinct from proprietary tooling).

Diverse Programming Environment

Diverse Programming Environment



- Organizations may choose:
 - R-only
 - SAS-only
 - SAS & R hybrid
 - other
- PharmaForest does not prescribe a path; it expands options for SAS or hybrid SAS–R organizations.
- Python is often used for AI/ML; multi-language environments are common.
- Risk/challenge: skill variability across languages can create inefficiencies; industry alignment matters.

Conclusion

Conclusion



- Open-source collaboration is reshaping both R and SAS communities.
- PharmaForest bridges “tradition and innovation” by bringing community-driven sharing into SAS.
- Long-term vision: R, SAS, and open source coexist and complement each other in pharmaceutical analytics.

Thank you!!



Join the community and be a **forester!**
We look forward to your participation.

Join PharmaForest:

<https://forms.office.com/r/UVPyB16J2U>



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Q & A

